

West Hartford Deck BOM v19

West Hartford Deck — Bill of Materials

Date: 2026-05-11 **Prepared by:** Hollis, Home-Improvement Concierge / Owner's Advocate **Rev:** v19 (2026-05-11)

Scope

23'-6" × 16'-3" attached deck on existing concrete patio, West Hartford CT. Maximum 12" above finished grade. Attached to wood-frame house via 2×10 PT ledger on the 23'-6" long (house-parallel) edge — **note: ledger is 20'-0" long; see Owner's Note A for the supplemental-support requirement on the remaining ~3'-0" of the house-side edge.** Two doubled 2×10 PT girder lines parallel to house, each on 5 Pylex 50 screw piers — mid-girder at y=8'-0", outer-rim girder at y=16'-0". Face-hung 2×10 PT joists at 12" o.c., perpendicular to house, each spanning in two segments (ledger to mid-girder; mid-girder to outer-rim girder). Center joist at x=11'-6" framing centerline supports center divider breaker board. MoistureShield Vision composite decking parallel to house in two 11'-9" halves with center divider breaker, 4-sided picture frame (all four sides continuous), composite fascia on 3 exposed sides. All hardware ACQ-compatible (hot-dip galvanized or stainless).

1. Foundation & Piers

Item	Spec	Qty	Notes
Pylex 50 helical/screw pier	50" galvanized shaft, ≥ 3" helix	10	5 per girder line × 2 girder lines; see pier layout note below
Pylex saddle bracket	Pylex 10555 (3.5" wide saddle for doubled 2×10)	10	Bolts to shaft top; doubled girder bears in saddle
1/4" cedar or HDPE rot-resistant shims	Min. 1.5" × 3.5" × 1/4" shim stock	40	2 shims per saddle × 2 sides × 10 saddles = 40 total. Use cedar or HDPE only — no OSB, no wood composite, no plywood
1/2" HDG carriage bolt + nut + washer (saddle-to-shaft)	1/2" × 3 1/2" HDG	20	2 per saddle × 10 saddles (per Pylex install spec)
	1/2" × 4" HDG carriage bolt	20	2 per saddle × 10 saddles. Through-bolted: drill through

Item	Spec	Qty	Notes
1/2" × 4" HDG carriage bolt (girder-to-saddle, through-bolt)			full doubled girder (3" of wood) plus saddle plate; carriage bolt from below with square shoulder seated in saddle; washer + nut on top of girder. Verify saddle plate thickness — use 4" minimum; 4½" if plate is thicker than ¼"
1/2" HDG flat washer (girder-to-saddle)	HDG flat washer, 1/2"	20	1 per carriage bolt, bearing on top of girder
1/2" HDG hex nut (girder-to-saddle)	HDG hex nut, 1/2"	20	1 per carriage bolt
Hydraulic cement or non-shrink grout	10-lb bag	3	Grout collar at patio surface at each pier shaft; 10 holes through patio slab
Concrete patio repair mix	Quikrete Vinyl Concrete Patcher or equiv., 10-lb tub	2	Cosmetic flush-fill around shaft collars at 10 locations

Pier layout: 10 locations in 2 rows of 5. - **Mid-girder line at y=8'-0"**: x = 0', 5'-9", 11'-6", 17'-3", 23'-0" - **Outer-rim girder line at y=16'-0"**: x = 0', 5'-9", 11'-6", 17'-3", 23'-0"

Pier spacing along each girder line is **5'-9"** (four bays). DCA-6 Table 3A: doubled 2×10 SP at 8-ft joist span = 8'-2" maximum allowable girder span. Actual pier spacing 5'-9" — passes with **29% margin**. This layout provides 4-span continuous beam behavior, distributing load more favorably than a simply-supported configuration.

Sledgehammer-punch 3–4" diameter holes through the existing patio slab at all 10 locations before driving piers. Confirm all 10 marks fall within the 23'-6" × 16'-6" patio footprint before punching.

Girder-to-saddle connection — carriage bolts vs. lags: The saddle-to-girder connection uses 1/2" × 4" HDG carriage bolts through-bolted through the doubled girder and saddle plate, rather than withdrawal-loaded lag bolts. A full through-bolt is a stronger connection than a lag driven into end/bottom grain — it distributes shear across the full bolt cross-section. Drill through both plies of the doubled 2×10 (3" of wood) plus the saddle plate, seat the carriage bolt from below with the square shoulder bearing in the saddle hole, and torque a flat washer + hex nut on top. Note: the saddle-to-shaft carriage bolts (attaching the saddle plate to the Pylex shaft) are a separate item specified immediately above and are unaffected by this detail.

2. Lumber — Pressure-Treated SYP, Ground-Contact (UC4B)

All PT lumber: Southern Yellow Pine, ground-contact rated (UC4B), KDAT (kiln-dried after treatment) preferred. Verify net dressed length at the yard before delivery.

2a. Ledger

Item	Size	Length	Qty	Notes
Ledger board	2×10 PT UC4B	20'-0"	1	Order a standard 20-ft board — no field cut to length required. See Owner's Note A regarding the ~3'-0" house-side gap beyond ledger end.

2b. Girders (Doubled 2×10 PT)

Girders are **two lines running parallel to the house (x-direction)**, each a doubled 2×10 PT UC4B spanning 23'-0" between pier inboards.

The fundamental rule: butt joints must land directly over a pier center. A joint that falls between piers is a mid-span splice — never permitted in a two-ply built-up girder. With piers at $x=0'$, 5'-9", 11'-6", 17'-3", and 23'-0", the splice recipe below is derived from the actual pier locations.

Splice scheme (per girder line, 2 plies):

Ply	Pieces	Cut lengths	Joint locations	Source stock
Ply 1	2 pieces	11'-6" + 11'-6"	Butt joint over center pier at $x=11'-6"$	2 × 12-ft 2×10 PT (6" waste each; drops usable for blocking)
Ply 2	3 pieces	5'-9" + 11'-6" + 5'-9"	Joints over piers at $x=5'-9"$ and $x=17'-3"$	1 × 12-ft 2×10 PT (center piece) + 2 × 8-ft 2×10 PT (end pieces, 2¼" waste each)

Ply 1 and Ply 2 joints are staggered to different piers — joints never occur at the same cross-section in both plies simultaneously. This is the mandatory stagger rule; do not deviate.

Board count per girder line (1 girder, 2 plies): - Ply 1: 2 × 12-ft boards - Ply 2: 1 × 12-ft board + 2 × 8-ft boards - **Total per girder line: 3 × 12-ft + 2 × 8-ft**

Board count for 2 girder lines (no spare): 6 × 12-ft + 4 × 8-ft

Order with 1 spare board each length: 7 × 12-ft 2×10 PT UC4B + 5 × 8-ft 2×10 PT UC4B for girders

Item	Size	Length	Qty	Notes
Girder ply — 12-ft piece	2×10 PT UC4B	12-ft	6	2 girder lines × 3 pieces per line (2 for Ply 1 + 1 for Ply 2 center). Field-cut Ply 1 pieces to 11'-6" (6" waste per board — save drops for blocking)
Girder ply — 8-ft piece	2×10 PT UC4B	8-ft	4	2 girder lines × 2 pieces per line (Ply 2 end pieces, each cut to 5'-9" with 2¼" waste)
Spare girder ply — 12-ft	2×10 PT UC4B	12-ft	1	1 spare board
Spare girder ply — 8-ft	2×10 PT UC4B	8-ft	1	1 spare board

Order: 7 × 12-ft 2×10 PT UC4B + 5 × 8-ft 2×10 PT UC4B for girders.

Splice rule (mandatory): Butt joints must land **directly over a pier center**, never midspan. Stagger plies so joints fall on different piers: Ply 1 joint at x=11'-6" (center pier); Ply 2 joints at x=5'-9" and x=17'-3" (piers 2 and 4). This stagger ensures the doubled girder is never both plies unsupported at the same cross-section.

Saddle-fit rationale: A doubled 2×10 (2 × 1.5" actual = 3.0" total width) fits inside the Pylex 10555 saddle (3.5" interior) with ¼" of clearance per side — shimmed plumb with the ¼" cedar/HDPE shims specified in Section 1.

Structural note: 5 piers create a 4-span continuous beam. The maximum bending moment in a continuous multi-span beam is lower per span than in a simply-supported configuration. This framing is structurally generous.

2c. Joists (Face-Hung, Two Segments Per Row)

Joists run **y-direction (perpendicular to house)**, spanning the 16'-0" deck depth in two equal segments split at the mid-girder.

Item	Size	Length	Qty	Notes
	2×10 PT UC4B	~7'-11¼" clear span each (order 8-ft)	48	24 regular rows × 2 segments each. Each

Item	Size	Length	Qty	Notes
Joist segments (regular rows, both halves)				segment spans ~7'-11¼" clear (ledger face to mid-girder face for inboard segment; mid-girder far face to outer-rim girder face for outboard segment). Order 8-ft, field-cut to fit
Center joist segments (extra row at x=11'-6" framing)	2×10 PT UC4B	~7'-11¼" each (order 8-ft)	2	1 extra center joist row × 2 segments; provides bearing for the center divider breaker board directly beneath it

Total joist segments: 50 (25 rows × 2 segments).

Clear span calculation: Each 8-ft segment spans from one framing face to the next. Mid-girder doubled 2×10 actual width = 3.0"; outer-rim girder doubled 2×10 actual width = 3.0". Inboard bay clear: ledger face to mid-girder house-side face ≈ 7'-11¼". Outboard bay clear: mid-girder far face to outer-rim girder inboard face ≈ 7'-11¼". IRC R507 prescriptive span table: 2×10 SYP at 12" o.c. supports 10'-4" to 13'-0" + depending on load category — a 7'-11¼" span has substantial margin.

Order: 55 × 8-ft 2×10 PT UC4B for joists (50 segments needed + ~10% waste/defect).

2d. Blocking

Blocking runs x-direction (perpendicular to joists). Two blocking rows recommended to stiffen mid-span.

Item	Size	Length	Qty	Notes
Field blocking (lateral stiffness)	2×10 PT UC4B	~10¼" (joist bay width at 12" o.c.)	~46	Two blocking rows at y=4'-0" (mid-span of inboard bay) and y=12'-0" (mid-span of outboard bay). Each row spans 23'-0" with one block per joist bay. 23 joist bays × 2 rows = 46 blocks. Cut from 8-ft drops from joist material

Item	Size	Length	Qty	Notes
House-side PF backing blocks	2×10 PT UC4B	~9" each	~25	Short pieces at 12" o.c. between ledger face and first joist face; back the house-side picture-frame board

Tip: Field blocking cuts from joist drop-offs (8-ft stock yields one segment + 4-ft drop → 4 blocking cuts). Order 4 extra 8-ft 2×10s beyond joist count if using 8-ft blocking-dedicated stock preferred.

2e. Lumber Order Summary

Category	Stock Size	Qty to Order
Ledger	2×10 PT 20-ft	1
Girder plies	2×10 PT 12-ft	7 (6 needed + 1 spare)
Girder plies	2×10 PT 8-ft	5 (4 needed + 1 spare)
Joists	2×10 PT 8-ft	55 (50 needed + ~10% waste)
Blocking (field + house-side)	2×10 PT cuts from drops	— (cut from joist drop-offs; order 4 extra 8-ft boards if blocking-dedicated stock preferred)

Note on girder stock: 12-ft boards are the girder ply stock. 16-ft boards are not needed. Confirm 12-ft UC4B is available in quantity at the yard.

3. Decking, Picture Frame & Fascia

All MoistureShield Vision composite, capped, DiamondDefense scratch coat, 50-yr fade & stain warranty. Available at Home Depot, Lowe's, Richards Building Supply (Hartford metro).

3a. Field Planks

Field decking runs **x-direction (parallel to house)** in two halves of equal width, each half spanning 11'-9" from the left or right PF inboard edge to the center breaker board.

Item	Brand / Product	Width	Thickness	Cut Length	Qty	Notes
Field decking	MoistureShield Vision	5.4"	1.0"	11'-9" (from 12-ft stock)	32	16 plank rows left of center

Item	Brand / Product	Width	Thickness	Cut Length	Qty	Notes
planks — left half						× 1 piece each; butts at center divider breaker
Field decking planks — right half	MoistureShield Vision	5.4"	1.0"	11'-9" (from 12-ft stock)	32	16 plank rows right of center × 1 piece each; butts at center divider breaker
Field planks (ordered)	MoistureShield Vision 12-ft	—	—	12-ft boards, field-cut to 11'-9"	72	64 pieces needed + 10% waste = 70.4 → order 72 × 12-ft Vision boards

Stock option note: 20-ft Vision boards can be cut to two 11'-9" pieces with only 6" waste per board, yielding 36 × 20-ft instead of 72 × 12-ft. Confirm 12-ft availability at Richards Building Supply (West Hartford) before ordering. If 12-ft is in stock, prefer it.

3b. Picture-Frame Boards (4 Sides plus Center Divider)

Side	Length	Ordered As	Qty	Corner Treatment
House-side PF	23'-6"	24-ft board, field-cut; or 2× 12-ft spliced over a joist	1	Butt at left/right against house wall
Far-side PF	23'-6"	Same	1	45° mitered at left/right outside corners
Left-side PF	16'-3"	18-ft board field-cut to 16'-3"; or splice 2 shorter boards over a joist	1	Mitered at far corner; butt at house end. Continuous — no stair interruption

Side	Length	Ordered As	Qty	Corner Treatment
Right-side PF	16'-3"	Same	1	Mitered at far corner; butt at house end
Center divider breaker board	16'-0"	Single 16-ft or 20-ft board field-cut to 16'-0"	1	Face-screwed into center joist top; no miter; butts house-side PF on one end and far-side PF on the other

Center divider detail: The breaker board sits at **x=11'-9" finish position**. The center framing joist beneath it is at **x=11'-6" framing centerline**. The breaker is face-screwed to the center joist top with Cortex screws + Vision plugs at approximately 12" o.c. — 16 screws + 16 plugs for the 16'-0" run.

3c. Fascia Boards (3 Exposed Sides)

MoistureShield Vision composite fascia: 0.67" thick, 11.25" tall.

Side	Length	Ordered As	Qty	Notes
Far-side fascia	23'-0"	24-ft board field-cut; or 2× 12-ft	1	Full run; backs far-side PF
Left-side fascia	16'-0"	16-ft board, use full length	1	Full run
Right-side fascia	16'-0"	16-ft board, use full length	1	Full run

3d. Decking Board Order Summary

Product	Length	Qty to Order
MoistureShield Vision field planks	12-ft	72
MoistureShield Vision center divider breaker	16-ft or 20-ft (1 piece)	1
MoistureShield Vision PF boards	24-ft (house-side + far-side)	2
MoistureShield Vision PF boards	18-ft (left-side + right-side, 16'-3")	2
MoistureShield Vision fascia	24-ft (far-side)	1
MoistureShield Vision fascia	16-ft (left-side + right-side — exact length)	2

4. Hardware & Fasteners

All connectors and fasteners: hot-dip galvanized (HDG), ACQ-compatible. No electroplated zinc. Simpson Strong-Tie specified throughout.

4a. Simpson Connectors

Item	Model	Qty	Notes
Girder end-to-ledger hanger (doubled 2×10)	Simpson HU210-2 (2-ply, HDG)	2	One at each end of the mid-girder line at the ledger (x=0' and x=23'-0") — outer-rim girder does not connect to ledger
Joist-to-ledger face-mount hanger	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each at ledger end
Joist-to-mid-girder face-mount hanger (house-side face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each hanging from mid-girder house-side face
Joist-to-mid-girder face-mount hanger (far-side face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each hanging from mid-girder far face
Joist-to-outer-rim-girder face-mount hanger (inboard face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each; joists terminate at outer-rim girder inboard face
Post base / saddle (pier)	Pylex 10555 saddle bracket	10	Already counted under Section 1
Deck-to-house lateral connector	Simpson DTT2Z (or MTWDeck equivalent)	4	IRC R507.9 minimum; deck is 16'-3" wide. Space evenly along ledger, avoiding HU210-2 hanger cluster zones

Hanger count total — LUS210Z: 25 (ledger) + 25 (mid-girder house face) + 25 (mid-girder far face) + 25 (outer-rim girder) = **100 LUS210Z**. Add 5% overage: **order 106 LUS210Z** (round up to 110 for convenience).

4b. Ledger-to-House Fasteners

Item	Spec	Qty	Notes
Ledger lag bolts (field, between hanger locations)	1/2" × 4" HDG lag	34	

Item	Spec	Qty	Notes
			1/2" HDG lag at 18" o.c. two-row staggered between hanger-connection zones
Through-bolts at girder-to-ledger (2 locations)	1/2" × 6" HDG carriage bolt + nut + 2 washers	12	6 per cluster × 2 cluster locations (x=0', x=23'-0') at mid-girder line end-to-ledger connections
Ledger through-bolt backup blocks (interior band joist)	2×10 PT scrap	2 short pieces	Back-blocking behind ledger at each carriage-bolt cluster

4c. Girder Assembly (Doubled 2-Ply)

Item	Spec	Qty	Notes
Ply-to-ply structural screws	GRK RSS or Simpson SDS 1/4"×3" HDG	~96	2 girder lines × 23'-0" run × ~2 screws per lineal foot = ~92; round to 96
Alternate: 1/2" carriage bolt + nut + washer (AWC spec for 2-ply girder)	1/2" × 3 1/2" HDG	~48	~1 per 24" o.c. × 46' total girder run = ~23; double for 2-ply bolt-through = ~46; round to 48

4d. Joist Hanger Fasteners

Item	Spec	Qty	Notes
Joist hanger nails/screws — LUS210Z	Simpson N10 (10d × 1 1/2") or SDS 1/4"×1 1/2" screws per Simpson spec sheet	~1,400	100 hangers × ~14 nail holes per LUS210Z = 1,400; buy 1,500
HU210-2 fasteners (girder-end-to-ledger)	Per Simpson spec (SDS 1/4"×1 1/2" or N10 as spec sheet requires for 2-ply)	~60	2 hangers × ~30 fastener holes each
DTT2Z structural screws	SDS 1/4"×4 1/2" per Simpson spec	~32	~8 per DTT2Z device × 4 devices

Critical: Verify exact nail/screw specification on the Simpson packaging for each hanger model. Do NOT substitute nail sizes.

4e. Decking Fasteners

Item	Brand / Spec	Qty	Notes
Hidden fasteners — field planks	Tiger Claw TC-G clips	~700	25 joist rows × 32 planks × ~16 clip points per plank = 512 clips per half × 2 halves; buy 700 with overage
#8 stainless screws for TC-G clips	Tiger Claw specified (stainless, 1-5/8")	~700	One per clip
Cortex face-screws + color-matched plugs — PF boards (4 sides + center divider)	Cortex for MoistureShield (must match Vision colorway)	5 boxes (100-ct each)	~4 screws per lineal foot × ~80 lf four-sided PF + 16 lf center divider = ~384 screws
Stainless trim screws — fascia to rim/drop-board	#10 × 2½" stainless	1 box (100-ct)	~3 screws per lineal foot × ~55 lf of fascia ≈ 165

5. Membrane & Sealants

Item	Brand / Spec	Coverage / Size	Qty	Notes
Peel-and-stick flashing tape (ledger)	Henry Blueskin VP160 or Carlisle CCW-705	6" wide roll	2 rolls (75 lf each)	Apply to house sheathing behind ledger before ledger install; 23'-0" run
Z-flashing, aluminum	2" leg × 2" leg, 10-ft sticks	3 sticks	Tucked behind siding/housewrap above top of ledger	
Kick-out flashing	Aluminum or plastic, pre-bent	2	At each end of ledger where	

Item	Brand / Spec	Coverage / Size	Qty	Notes
			siding terminates	
Joist / girder top tape	Vycor Deck Protector 2" or Trex Protect 2½"	2" wide roll, 50 lf per roll	11 rolls	Apply to top of every joist segment (50 segments × ~8 lf each ≈ 400 lf), both girder top runs (~46 lf each × 2 ≈ 92 lf), and ledger top (23 lf) = ~515 lf total
End-grain sealant	Copper Green Wood Preservative or Anchorseal 2	1-qt brush-on	2 qt	Seal ALL field cuts of PT lumber immediately after cutting
Backer rod + polyurethane sealant	Sikaflex 1A or similar, gray	1 tube	3 tubes	Ledger-to-house seam caulk
Ledger spacers / standoffs	Starborn Camo LedgerUp composite or PT shims	—	as needed	If siding extends behind ledger location
Drip cap (head flashing)	Aluminum drip cap, 2" face	24-lf sticks	1 stick	Over ledger at top if housewrap alone is insufficient

6. Lumber Order Summary

Category	Stock Size	Qty to Order
Ledger	2×10 PT 20-ft	1
Girder plies	2×10 PT 12-ft	7 (6 needed + 1 spare)

Category	Stock Size	Qty to Order
Girder plies	2×10 PT 8-ft	5 (4 needed + 1 spare)
Joists	2×10 PT 8-ft	55 (50 needed + ~10% waste/defect)
Blocking (field + house-side)	2×10 PT cuts from drops	— (cut from joist/girder drop-offs; add 4 extra 8-ft boards if dedicated blocking stock preferred)

7. Owner's Notes (Hollis)

A. 20-ft ledger — supplemental support required for the remaining ~3'-0" of the house-side edge. Thomas, this is the note I want you to see before anything gets ordered or permitted. The deck framing is 23'-0" wide, and the ledger is 20'-0" long. That leaves approximately 3'-0" of house-side edge — the last bay, roughly $x=20'-0"$ to $x=23'-0"$ — without a ledger to carry joist load back to the house. This is not a small detail. Before the permit is submitted, one of the following must be resolved:

- **(a) Supplemental short ledger segment:** A second ledger piece, roughly 3'-3" long, is installed and flashed to cover the remaining span to $x=23'-0"$. Requires flashing, lag or through-bolt connections per DCA-6 requirements, and a lap detail or gap detail between the two ledger segments. This is the cleanest solution.
- **(b) Beam-and-post arrangement:** The last bay of joists at the house side is carried by a drop beam and post(s) rather than a ledger, moving that corner load to the patio/grade rather than the house framing. Requires confirming post bearing on a footing — the existing pier at $x=23'-0"$ on the mid-girder line is at $y=8'-0"$, not $y=0'$; a separate bearing point near $y=0'$ at $x=23'-0"$ would be needed.
- **(c) Designer/inspector confirmation:** The designer or West Hartford building official confirms that the joist framing in that zone is otherwise supported by the deck structure alone without a full-width ledger. This is the least likely path to acceptance without engineering documentation.

I am spec'ing the 20-ft board as instructed. Do not proceed to permit submission or framing without resolving which of (a), (b), or (c) applies. Flag this for the plans examiner — they will catch it anyway.

B. DCA-6 Table 3A girder span — 29% margin. DCA-6 Table 3A, doubled 2×10 Southern Pine at 8-ft joist span column: maximum allowable girder span = 8'-2". Actual pier spacing in this design = 5'-9". That is 2'-5" under the DCA-6 limit — 29% margin. Print the DCA-6 table, circle the 8'-2" value, write "5'-9" actual" next to it, and put that sheet on top of your permit package. If the West Hartford plans examiner pushes back on the doubled girder or the parallel-to-house orientation, this number is your clean answer.

C. Splice rule — do not let the contractor improvise this. Girder ply joints must land directly over a pier center. In this design, that means Ply 1 joint at $x=11'-6"$ (center pier); Ply 2 joints at $x=5'-9"$

and $x=17'-3"$. These locations are non-negotiable. If a framer suggests splicing “wherever the board runs out,” that is a structural deficiency and a permit failure. The splice diagram in Section 2b is the spec. Write it on the framing plan and post it at the job site.

D. Saddle shim detail — 10 saddles, 40 shims. Each Pylex 10555 saddle gets a $\frac{1}{4}"$ cedar or HDPE shim on each side before the doubled girder drops in — 40 shims total. Pre-cut and label them before pier installation day. A framer who skips the shims and drives the saddle bolts against a rattling doubled 2×10 will create a racking condition that shows up as a squeak or a bounce after the decking is down. Do it right on the way up.

E. Saddle-to-girder carriage bolt connection. The girder-to-saddle connection uses $\frac{1}{2}" \times 4"$ HDG carriage bolts through-bolted through the doubled girder and saddle plate — 2 bolts per saddle, 20 bolts total — rather than withdrawal-loaded lag bolts. This is a stronger connection: a full through-bolt transfers shear in bearing across the bolt cross-section, where a lag bolt in the girder bottom face relies on withdrawal resistance alone. Spec the bolts at 4" minimum length (3" of wood + saddle plate + washer + nut clearance); use $4\frac{1}{2}"$ if the Pylex 10555 saddle plate is thicker than $\frac{1}{4}"$. Verify plate thickness before ordering. The saddle-to-shaft carriage bolts (attaching the saddle plate to the Pylex shaft itself) are a separate item — those stay as spec'd in Section 1.

F. Geometric alignment — center pier, center joist, center breaker. The five piers on each girder line are evenly spaced at $5'-9"$. The center pier falls at $x=11'-6"$. The center framing joist is at $x=11'-6"$. The center breaker board sits at $x=11'-9"$ finish ($x=11'-6"$ framing). These three elements — center pier, center joist, center breaker — all converge on the same x-line. The pier is on the girder, not on the joist; structurally they're different systems. But the deck's natural midpoint is reinforced at every layer from the foundation up. It's a byproduct of the arithmetic, not a thing we engineered, but it's a clean result.

G. Joist orientation — verify with the crew before framing. Joists run **perpendicular to the house** (into the yard). This is the reverse of the typical attached-deck framing instinct. Girders run parallel to the house; joists bridge between them into the yard. Verify with the crew foreman before framing begins.

H. Pier layout verification before punching holes. Before punching any holes, lay out all 10 marks with marking paint, string line, and a chalk snap. Verify each mark falls within the patio footprint ($23'-6" \times 16'-6"$) and at the correct x and y positions. Do not start punching until all 10 are verified.

I. Order PT lumber early. Kiln-dried-after-treatment (KDAT) is the right spec — drier, straighter, less post-install shrinkage. May need 1–2 weeks lead time at the yard. Call Richards Building Supply or Curtis Lumber (West Hartford area) early to confirm stock of KDAT SYP in the sizes needed. The girder stock is 12-ft boards — confirm 12-ft UC4B is available in quantity.

J. West Hartford building permit. This deck is ≥ 200 sq ft attached to the house — a building permit is required. Bring to the permit office: the plan drawings, this BOM, the Pylex 50 + Pylex 10555 saddle load data sheets, the DCA-6 Table 3A girder span check ($5'-9"$ actual vs. $8'-2"$ max = 29%

margin), the Simpson LUS210Z and HU210-2 spec sheets, and the Simpson DTT2Z lateral tie spec sheet. The contractor pulls the permit — not the homeowner.

K. No railing required by code. IRC R312 requires guardrail at decks $\geq 30"$ AG. This deck is $\leq 12"$ AG throughout. Confirm with the West Hartford building department at permit application.

L. End-grain sealing is not optional. Every cut end of every PT board must be sealed with Copper Green or Anchorseal 2 immediately after cutting. Keep the can at the saw station.

M. Cortex color — order after colorway is selected. Do not order Cortex plugs before the MoistureShield Vision colorway is finalized.

N. Joist tape coverage. ~515 lf total. Buy 11 $\times$ 50-lf rolls. Apply after framing inspection passes and before the first decking plank is set.

Inspection Sign-Off Block

Stage	Inspector	Signature	Date
Foundation / piers installed	West Hartford Building Dept.		
Framing complete (pre-deck)	West Hartford Building Dept.		
Ledger flashing	West Hartford Building Dept.		
Final inspection	West Hartford Building Dept.		

All inspections to be scheduled by the contractor. The contractor pulls the permit; the homeowner does not.

West Hartford Deck — Bill of Materials. Prepared by Hollis — Home-Improvement Concierge / Owner’s Advocate. Print this document and bring it to the lumber yard and hardware store. Quantities include waste factors where noted; consult your contractor or permit plans for final structural verification.